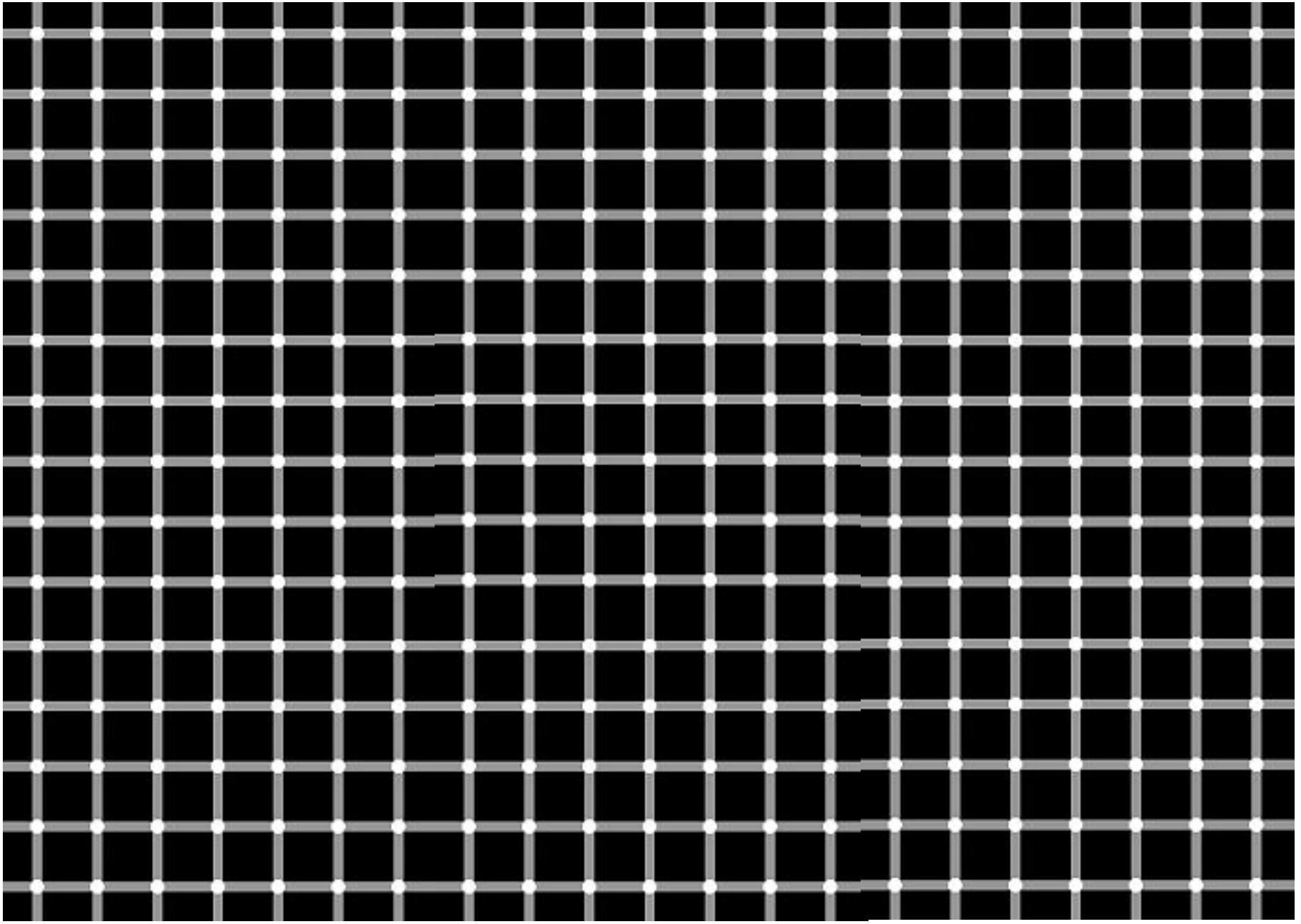


Self-Correcting Quantum Computers

Dave Bacon

Department of Computer Science & Engineering
University of Washington





funded by

Warning:
Lah-less Talk

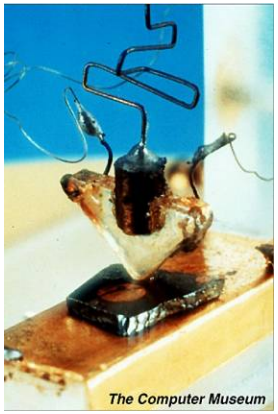
In This Talk We Will Obey The Lahs of Quantum Theory

It Was a Long Slog…



Alan Turing

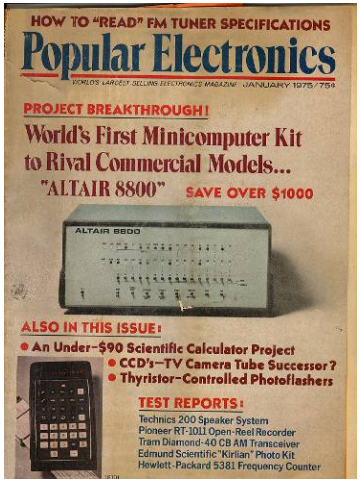
1936- Alan Turing: “On computable numbers, with an application to the Entscheidungsproblem”



1947- First transistor



1958- First integrated circuit



1975- Altair 8800



2005 GHz machines that weight ~ 1 pound

What is a Computer?

Practical question (as opposed to philosophical)

What is the phase of matter corresponding to the computer?

There are distinct physical reasons why robust classical computation is possible.

Hard Drive 

Integrated Circuit 

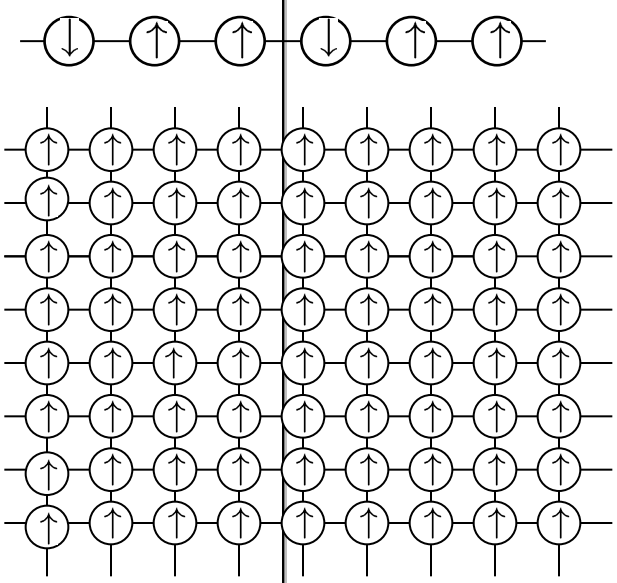
Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?

(1) The physics of classical robust computation

(2) An attempt to port these ideas to quantum computers

Three Storage Testbeds

Ferromagnetic Ising models:



one dimensional

two (and >) dimensional

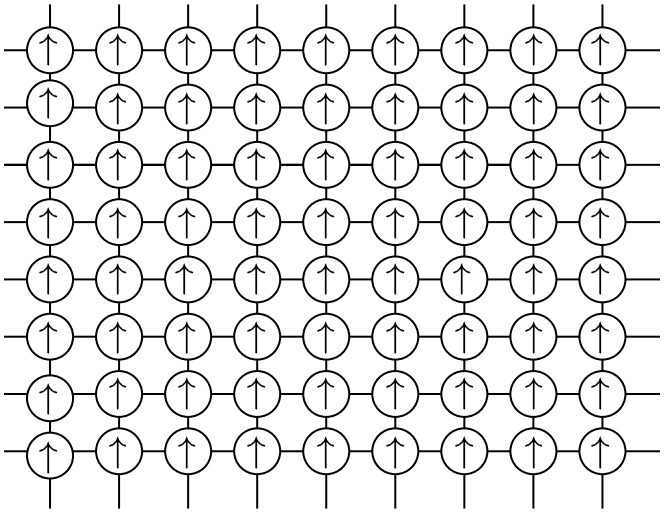
Toom's rule:

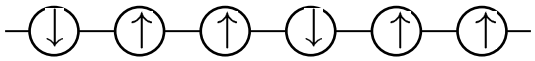
0	0	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	1	1	1	0
0	0	0	0	1	0	1	0
1	1	1	0	0	0	0	0

two dimensional noisy cellular automata

Ferromagnetic Ising Models







Ising model

spins on edges of a lattice: $s_i \in \{+1, -1\}$

energy of a configuration:

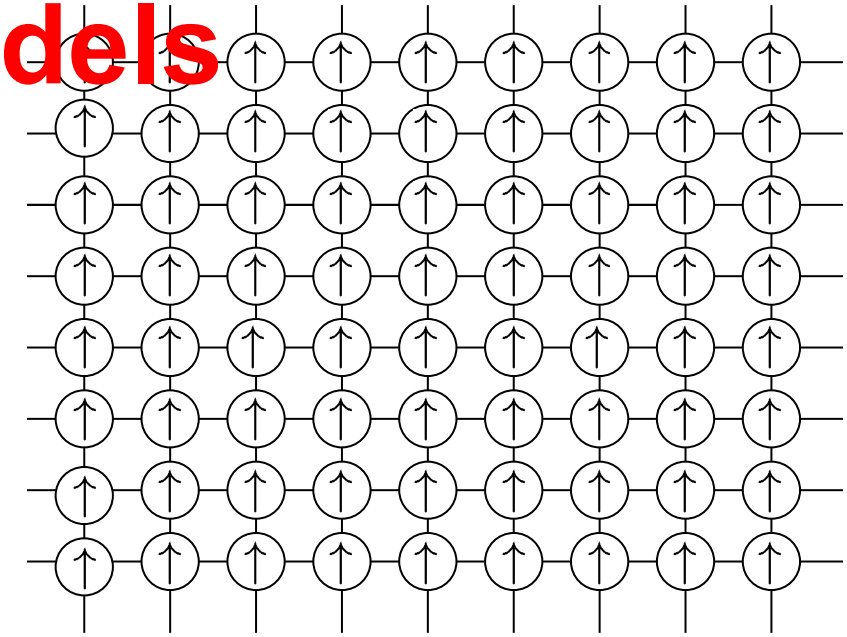
$$E = -J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i \qquad J > 0$$

sum over neighbors

external magnetic field

$h = 0$ for now

Dynamic Ising Models



Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

Update rule:

If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$



Reminder



two-level system perturbatively coupled to a thermal boson bath

$$\frac{\partial \boldsymbol{\rho}}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+ - 2\boldsymbol{\sigma}_+\boldsymbol{\rho}\boldsymbol{\sigma}_-\right]$$

"heating"

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\boldsymbol{\sigma}_+\boldsymbol{\sigma}_-\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_+\boldsymbol{\sigma}_- - 2\boldsymbol{\sigma}_-\boldsymbol{\rho}\boldsymbol{\sigma}_+\right]$$

"cooling"

$$-\phi\left[\boldsymbol{\rho} - \boldsymbol{\sigma}_z\boldsymbol{\rho}\boldsymbol{\sigma}_z\right]$$

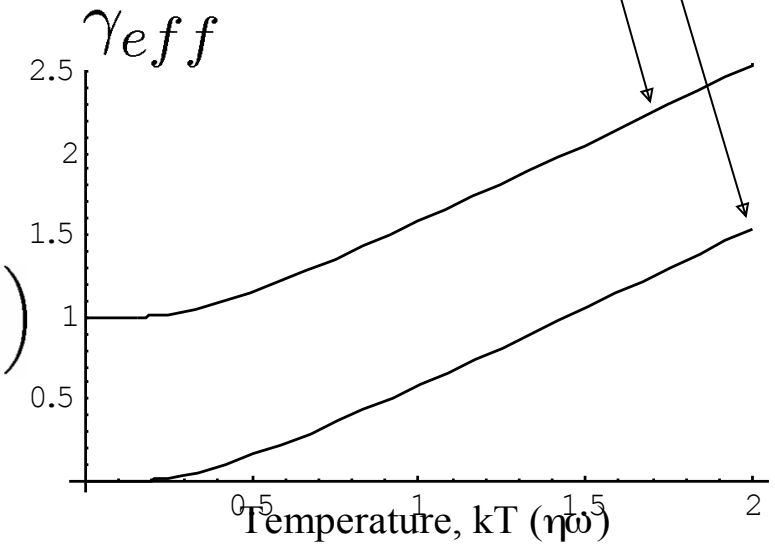
"non-dissipative"

$\hbar\omega$

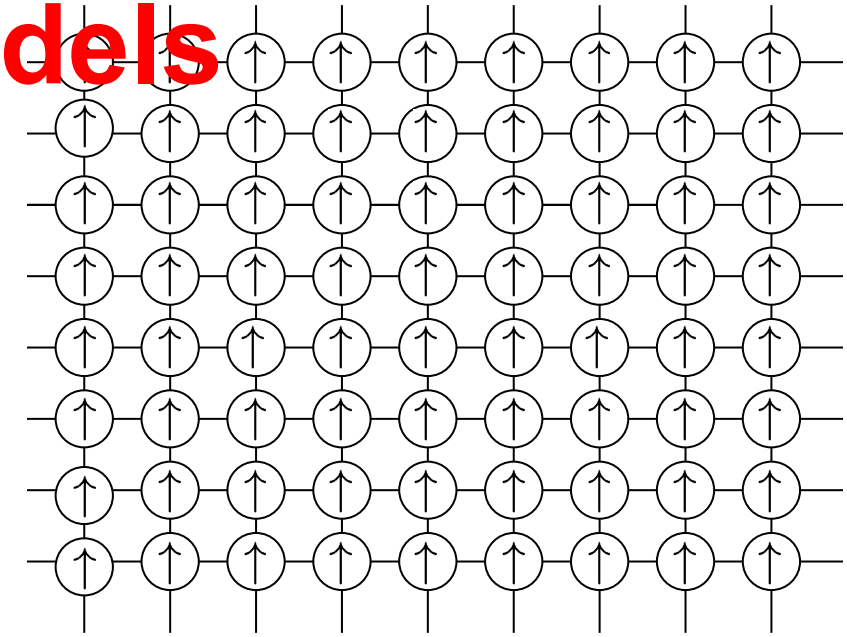
$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\hbar\omega}{kT}\right)+1}$$

$$\frac{\text{heating}}{\text{cooling}} = \frac{\bar{n}_{th}}{\bar{n}_{th}+1} = \exp\left(-\frac{\hbar\omega}{kT}\right)$$

$$= \exp\left(-\beta\Delta E\right)$$



Dynamic Ising Models



Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

Update rule (alternate with noise):

If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

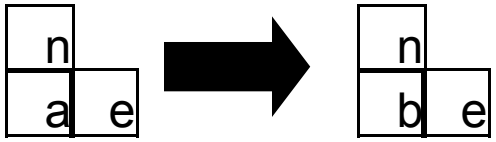
Toom's Rule



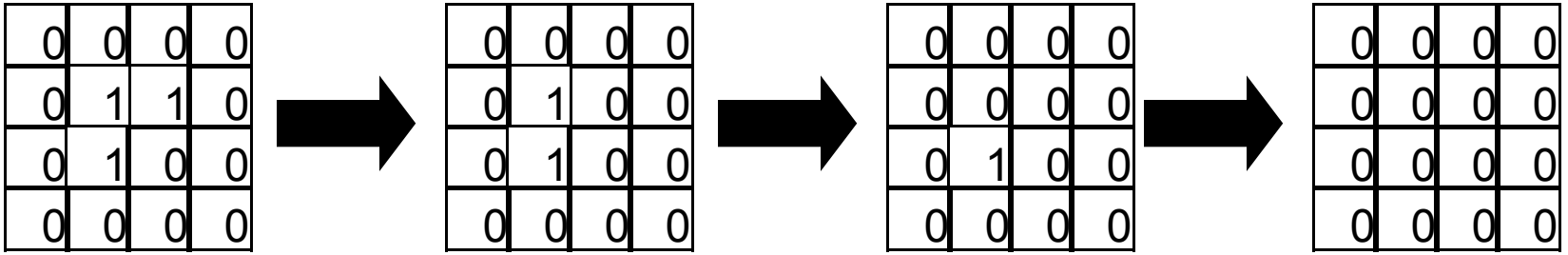
0	0	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	1	1	1	0
0	0	0	0	1	0	1	0
1	1	1	0	0	0	0	0

two dimensional noisy cellular automata

Update rule:



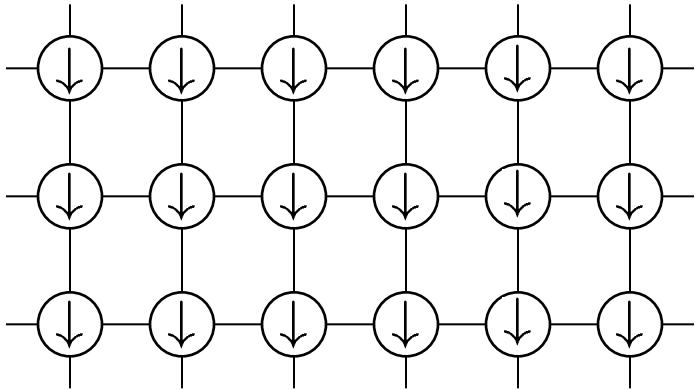
b=majority(n,e,b)



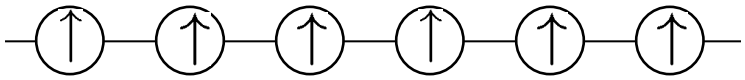
Update rule with noise: with some probability the update fails and a random state is assigned

Compare and Contrast

Encode information into the system and watch what happens



0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0



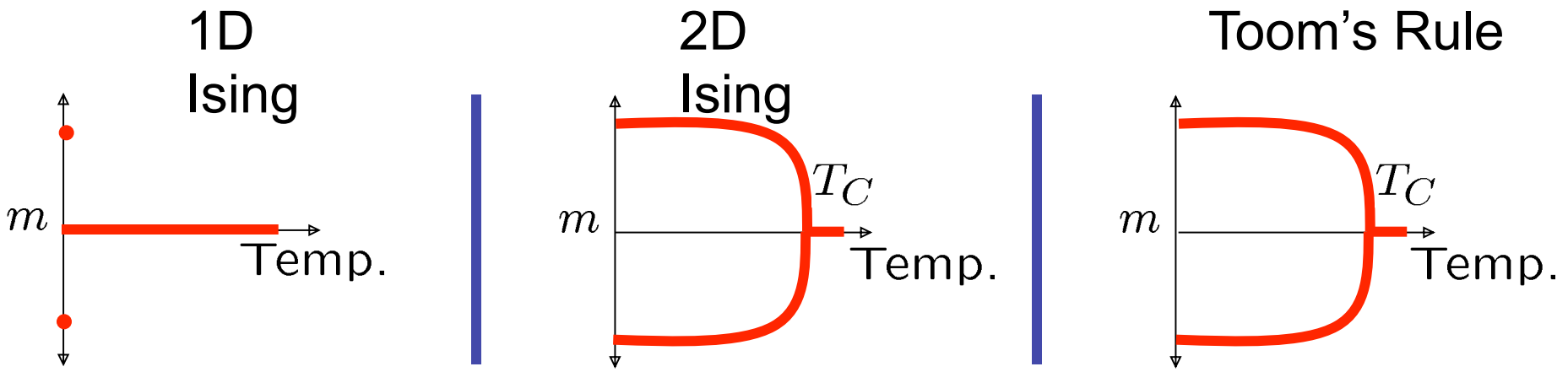
Redundancy code.

Order parameter is magnetization:

$$m = \frac{1}{N} \sum_i s_i$$

Storage

Long time behavior (not infinite time, but effectively)



Criticism: (1) thermodynamic limit taken

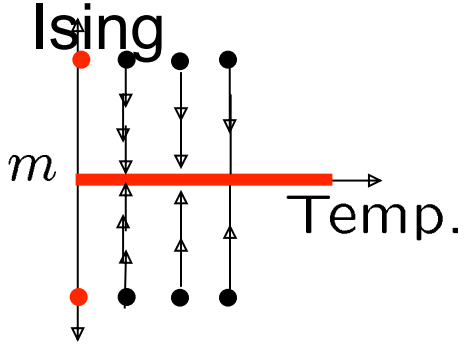
(2) what if relaxation to equilibrium takes a long time?

Storage



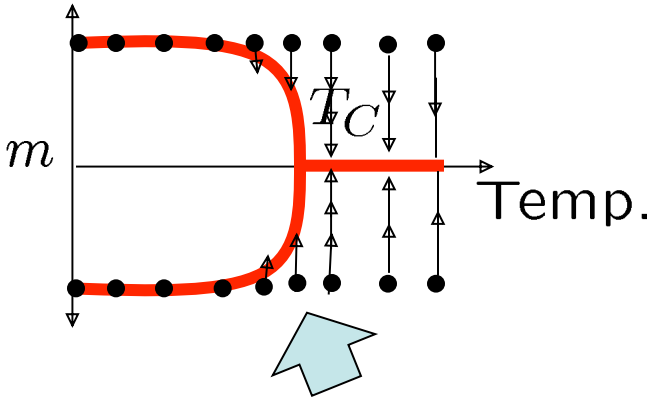
2nd attempt: 1D Ising

Relaxation to thermal



decay rate $\propto e^{-\beta J}$

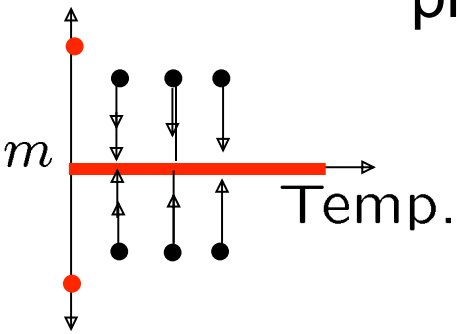
2D Ising, Toom's Rule

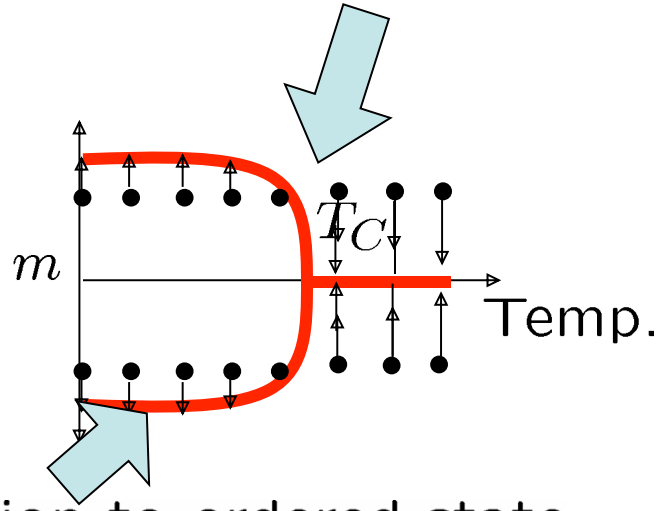


critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

Imperfect preparation





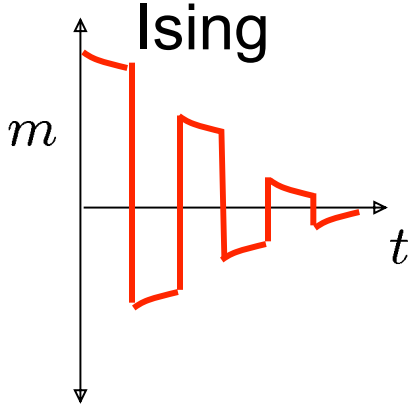
fast relaxation to ordered state

Manipulation

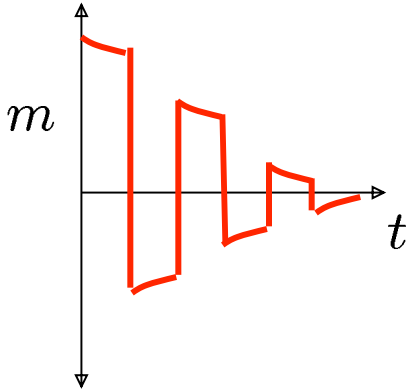


Flipping spins is imperfect

1D Ising

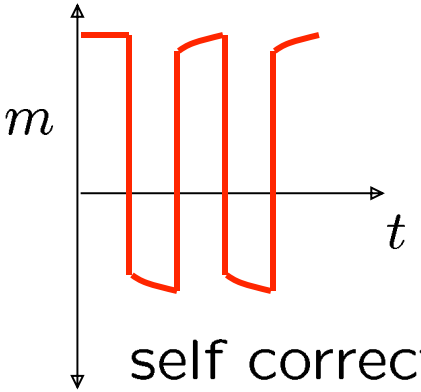


2D Ising Toom's Rule



$T > T_C$

1D Ising
- temperature less than gap (2J)
- manipulation error must be small (i.e. NOT fault-tolerant)

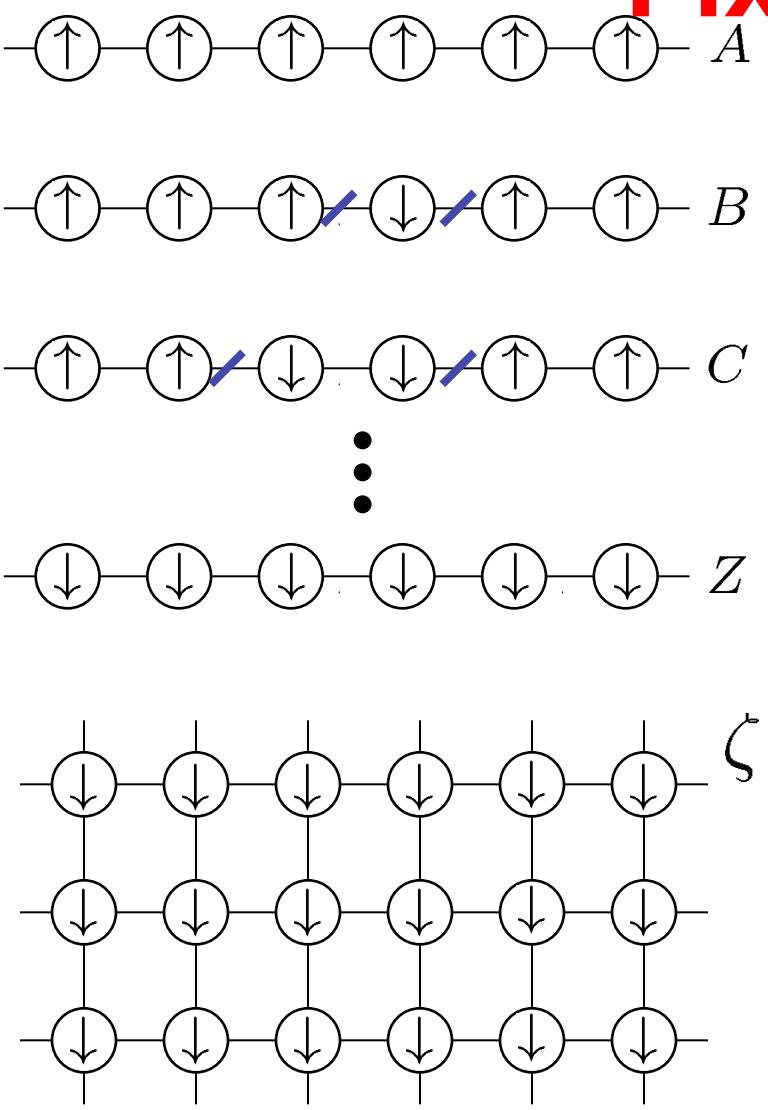


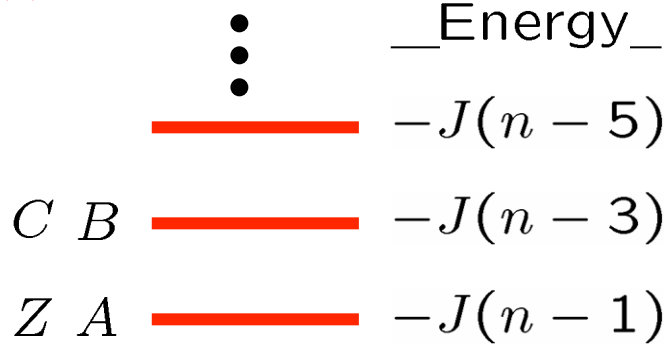
$T < T_C$

self correcting

2D Ising
- temperature less than critical temp

“Self Fixing”

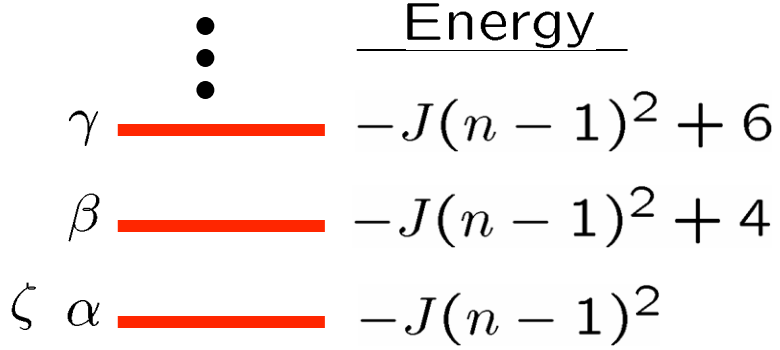




Excitations point-like

Energy to err is constant

Disordering entropy >> Energy



Excitations string-like

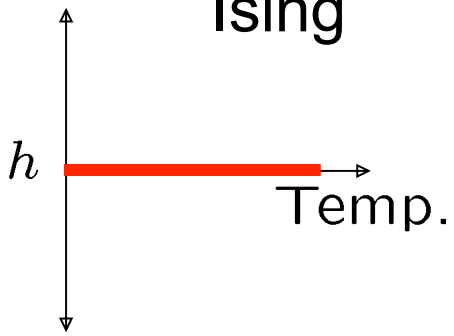
Energy to error is prop to size

Energy >> Disordering entropy

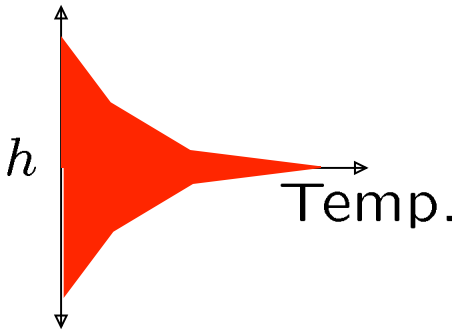
Toom's Rule Versus 2D Ising

Applied magnetic field (corresponds to a bias in noise)

2D Ising



Toom's Rule



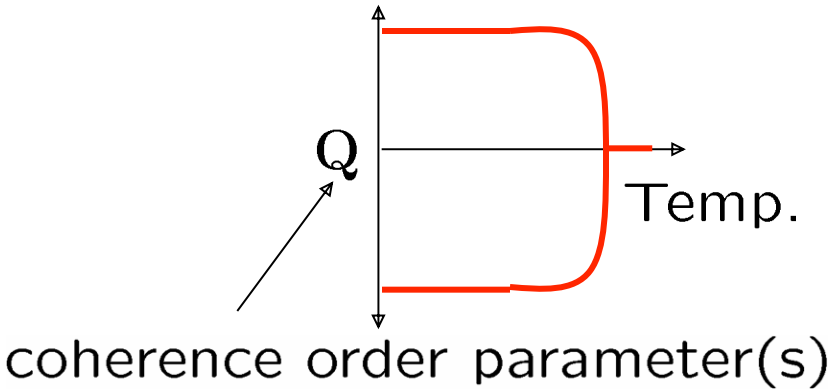
Regions with two coexisting phases

External magnetic fields disorder magnetic media

The Quantum Hard Drive?

(Kitaev)

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



1. Coherence preserving.

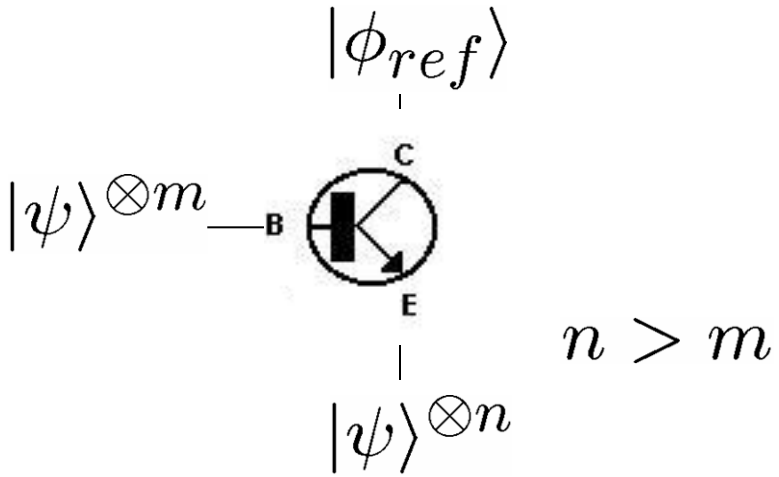
$Q_1 \Rightarrow -Q_1$

2. Accessible Fault-Tolerant Operations

$\langle G_1, \ldots, G_{n^k} \rangle \cong U(c^n)$

3. Universality

No Quantum Transistor?

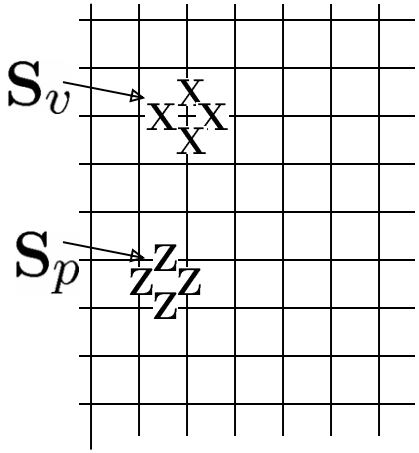




Quantum Cloning Machine

"A single quantum cannot be cloned," Wootters and Zurek, Nature, 1982

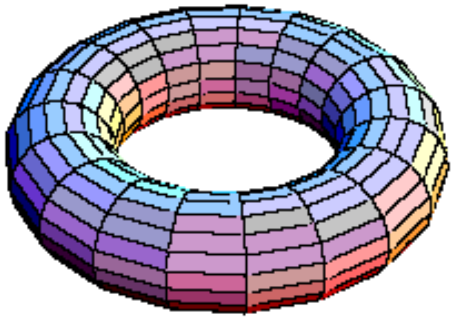
Kitaev's Toric Codes



Qubits on the edges of a lattice on a torus

(torus condition can be

Plaque operators:



$$S_p = \prod_{e \in p} Z_e$$

Vertex operators:

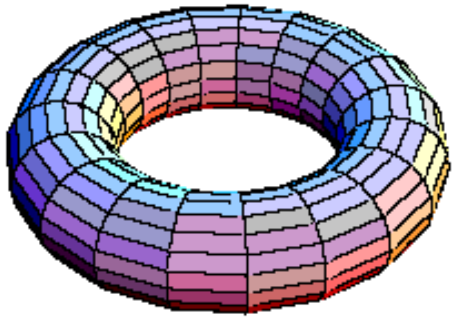
$$S_v = \prod_{e \in v} X_e$$

Stabilizer code: plaque and vertex opertors commute

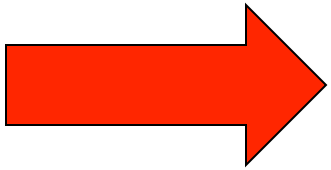
degenerate +1 eigenspace is quantum error correcting code

Kitaev's Toric C

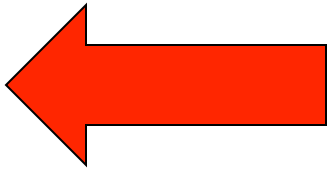
$$S_p = \prod_{e \in p} Z_e$$



+1	+1	+1	+1
+1	+1	+1	+1
+1	+1	+1	+1
+1	+1	+1	+1

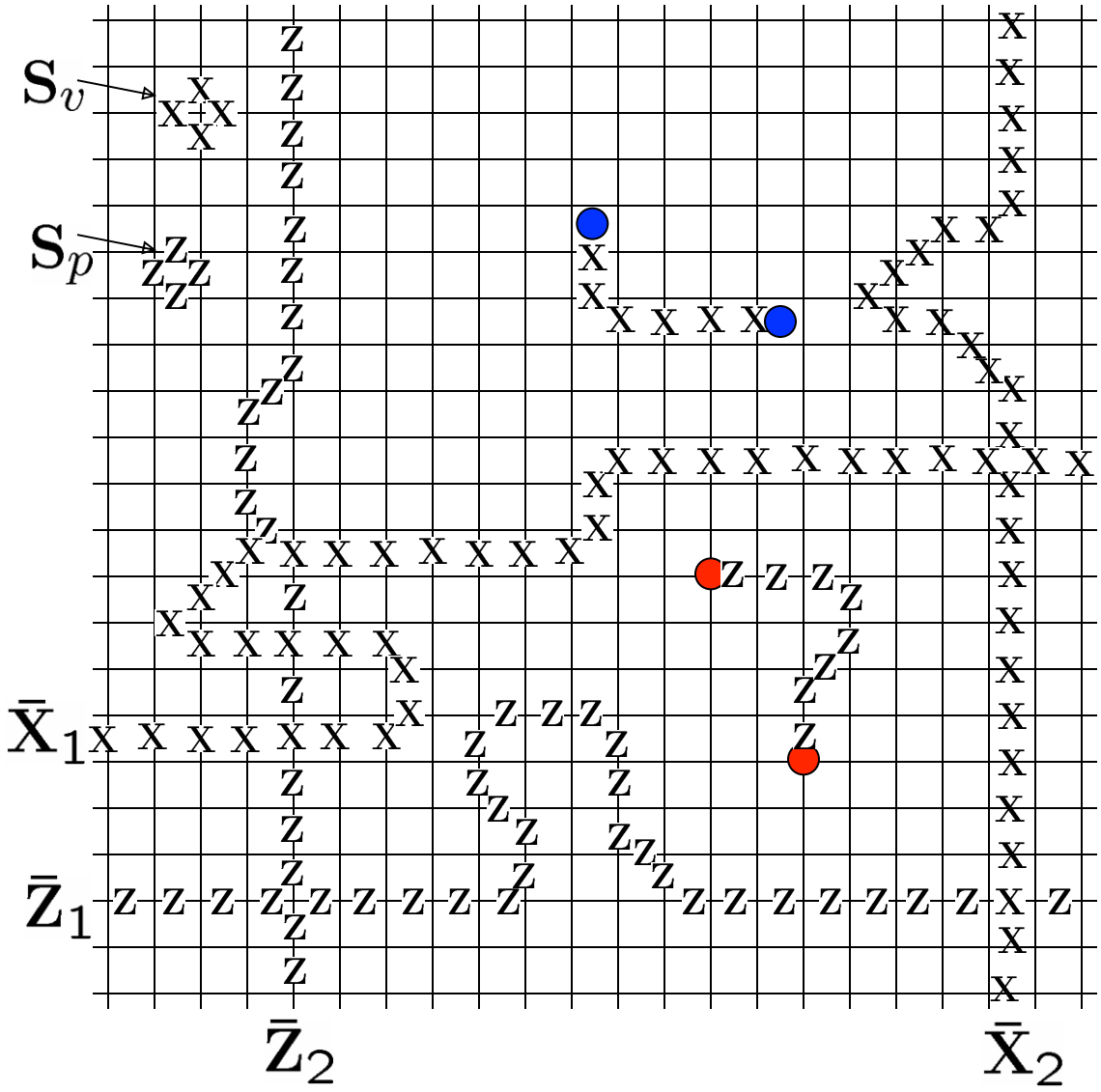


+1	-1 X	-1	+1
+1	+1	+1	+1
+1	+1	+1	+1



+1	+1	+1	+1
+1	-1 X X	+1	+1
+1	+1	-1	+1
+1	+1	+1	+1

Toric Code Correcting Properties

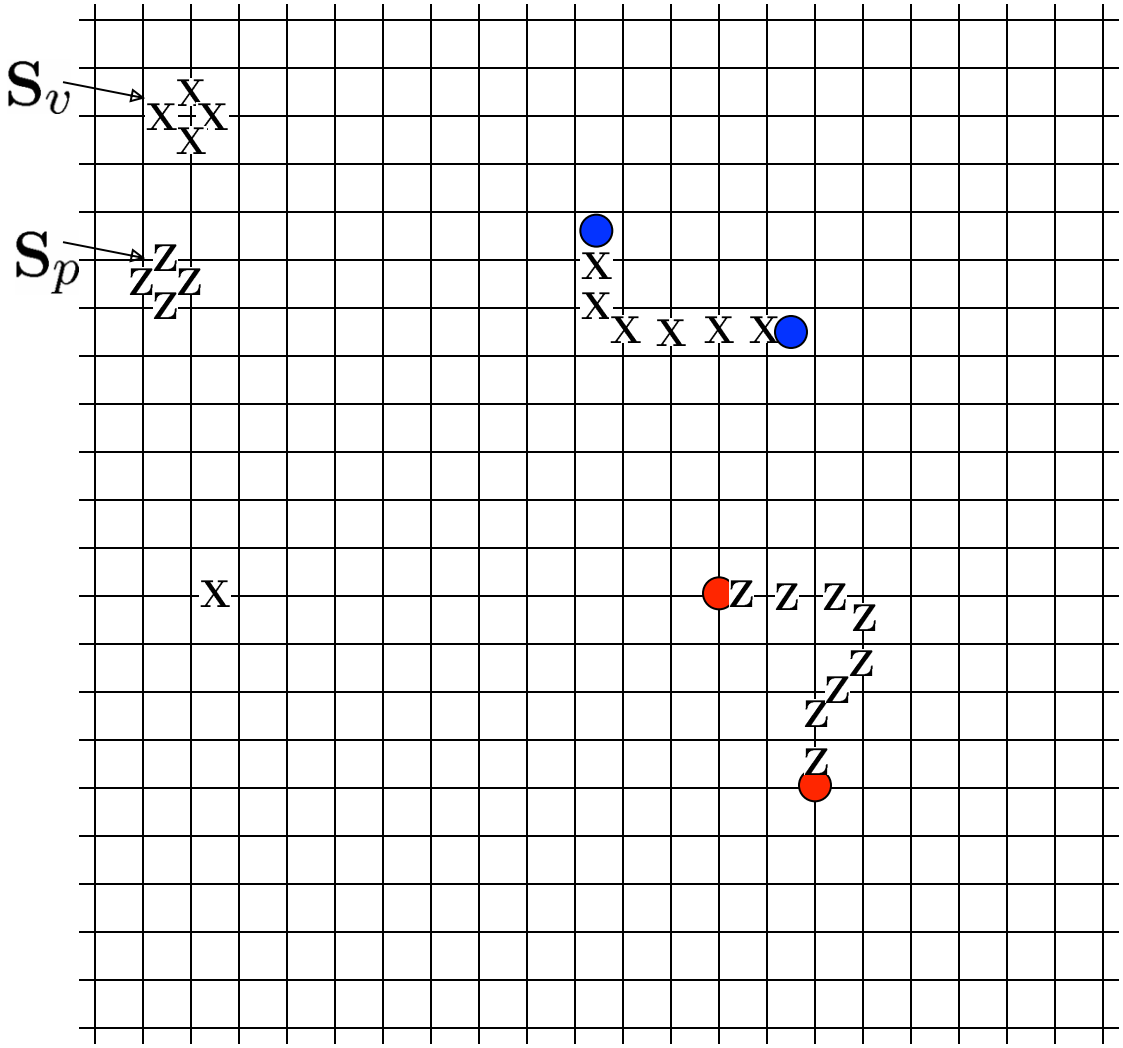


$$\mathbf{S}_p = \prod_{e \in p} \mathbf{Z}_e$$
$$\mathbf{S}_v = \prod_{e \in v} \mathbf{X}_e$$

Nonlocal matching algorithm allows the $O(L^2)$ode to correct qubit errors.

Toric codes provide fault-tolerant threshold for quantum computation with this nonlocal correcting algorithm.

Toric Code Hamiltonians



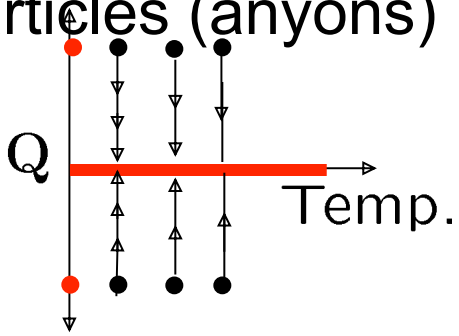
$$\mathbf{S}_p = \prod_{e \in p} \mathbf{Z}_e$$

$$\mathbf{S}_v = \prod_{e \in v} \mathbf{X}_e$$

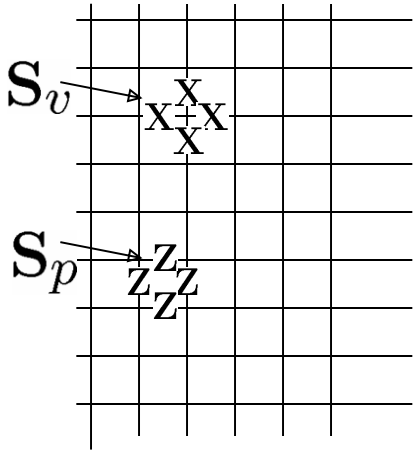
$$\mathbf{H} = -\frac{\Delta}{2}\left(\sum_p \mathbf{S}_p + \sum_v \mathbf{S}_v\right)$$

energy gap $= \Delta$

Excitations are point particles (anyons)



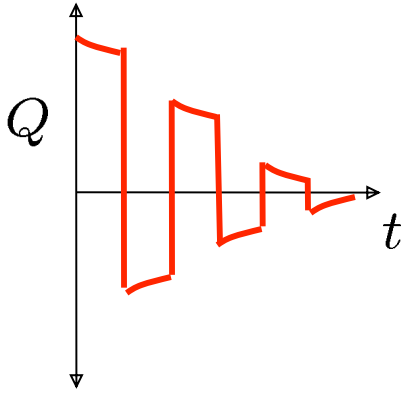
Kitaev's Toric Codes

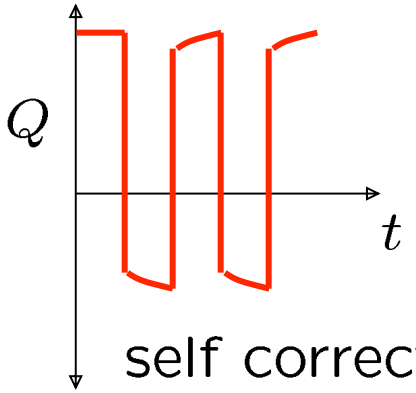


$$\mathbf{H} = -\frac{\Delta}{2}\left(\sum_p \mathbf{S}_p + \sum_v \mathbf{S}_v\right)$$

This Hamiltonian provides a mechanism for robust storage of quantum information.

But it does not provide a mechanism for robust manipulation of quantum information.

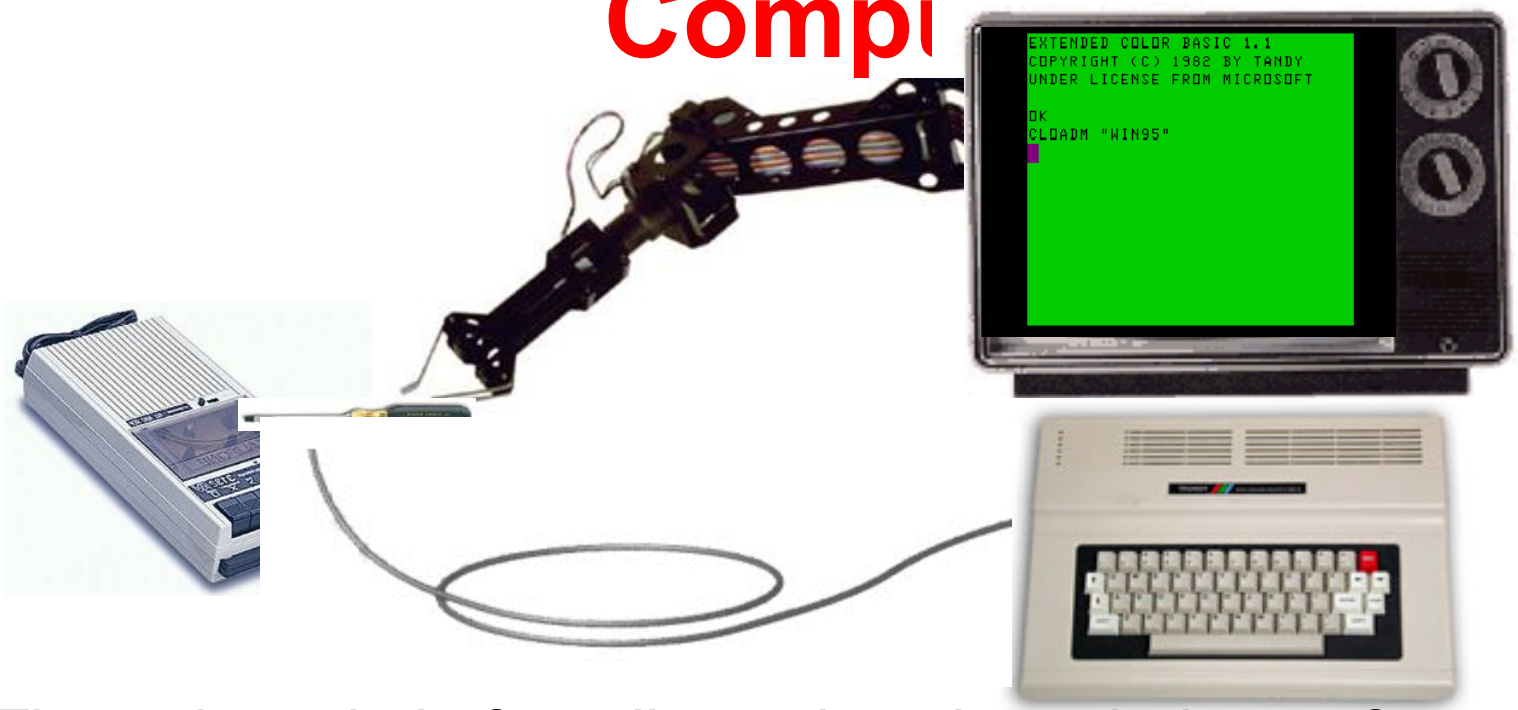




$T < T_C$

Is it possible to construct a Hamiltonian, which allows for fault-tolreant manipulation as well?

Self Correcting Quantum Compu



The toric code in four dimensions is probably self-correcting (two dimensions each for phase and bit flips)

Can we self-correct in realistic systems?

D. Bacon “Operator Quantum Error Correcting Subsystems for Self-Correcting Quantum Memories” quant-ph/0506023

Subspaces Versus Subsystems

The most general way to encode quantum information is into subsystem (not subspaces)



two

$\mathcal{H}_2 \otimes \mathcal{H}_2$

<u>encode a qubit in a subspace</u>

$$|0_L\rangle = |1\rangle \otimes |1\rangle$$

$$|1_L\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

$$\alpha|0_L\rangle + \beta|1_L\rangle$$

<u>encode a qubit in a subsystem</u>

$$(\alpha|0_L\rangle + \beta|1_L\rangle) \otimes |\psi\rangle$$

generic subsystem:

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

Quantum Error Correcting Subsystems

Knill and Laflamme, "A theory of quantum error correcting codes" PRA, 55, pp. 900-911 (1997)

Knill, Laflamme, and Viola, "Theory of quantum error correction for general noise," PRL, 84, pp. 2525-2528 (2000)

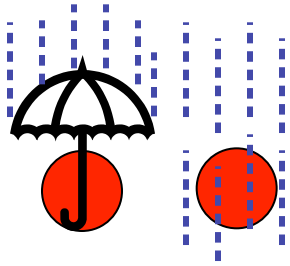
Subsystem encoding:

- implicit from the beginnings of quantum error correction
- explicit for degenerate (noiseless) codes and dynamic recoupling

Quantum error correcting subsystem codes:

Kribbs, Laflamme, and Poulin, "A Unified and Generalized Approach to Quantum Error Correction," PRL, 94, p. 180501 (2005)

Subsystem Error Correction



Only need to protect subsystem,
not full subspaces.

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

encoding subsystem

$$(|\phi\rangle\langle\phi| \otimes \rho_D) \oplus 0 \quad \mathcal{E} \quad \mathcal{R} \quad (|\phi\rangle\langle\phi| \otimes \tilde{\rho}_D) \oplus 0$$

encoded quantum information

error

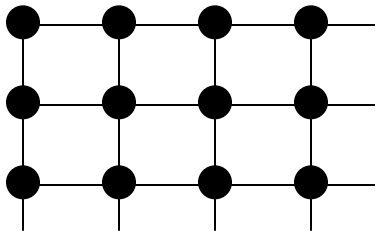
recovery

recovery need not perfectly restore system, but only preserve information encoded into subsystem.

Mysterious Extra Slide

Flatland

n by n cubic lattice of qubits:



$$X_{2,2} = \left[\begin{array}{cccc|cccc} 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & & \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{array}\right] \quad \begin{matrix} & & & & X & & & \\ & & & & & & & \end{matrix} \quad \begin{matrix} & & & & Z & & & \\ & & & & & & & \end{matrix}$$

$$Z_{2,1} = \left[\begin{array}{cccc|cccc} 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{array}\right]$$

$$Y_{2,1} = i \left[\begin{array}{cccc|cccc} 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{array}\right]$$

Generic Pauli group operator on cubic lattice:

$$P = i^k \left[\begin{array}{c|c} A & B \end{array}\right]$$ A, B n by n 0,1 matrices

X Columns, Z Rows

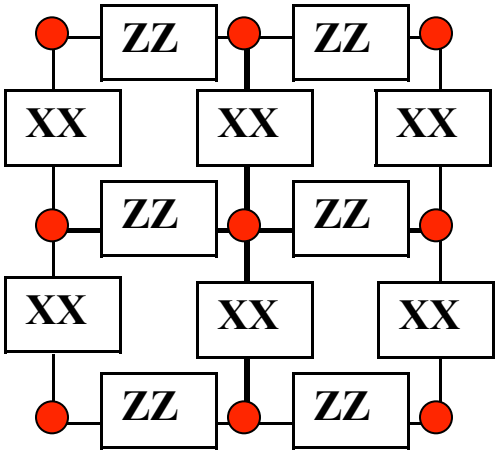
X Z

$$\left[\begin{array}{c|c} A & B \end{array}\right]$$

$\mathcal{T} = \{c\left[\begin{array}{c|c} A & B \end{array}\right]$ | A has an even # of 1′s in columns

B has an even # of 1′s in rows$\}$

$$c\left[\begin{array}{cccc|cccc} 0&1&0&0&1&1&1&1 \\ 0&1&0&0&0&0&0&0 \\ 0&0&0&1&0&0&0&0 \\ 0&0&0&1&1&1&0&0 \end{array}\right] \in \mathcal{T}$$

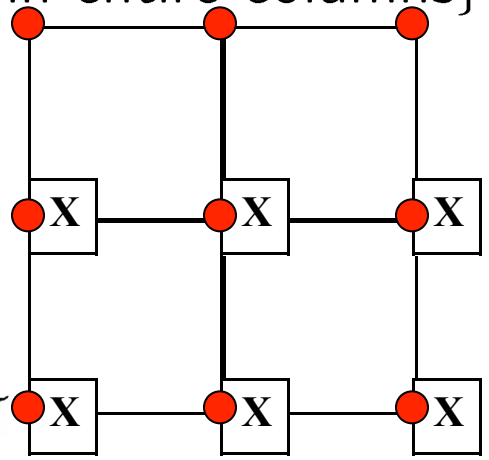


$\mathcal{S} = \{\left[\begin{array}{c|c} A & B \end{array}\right]$ | A has an even # rows with 1′s in entire row

B has an even # columns with 1′s in entire columns$\}$

$\mathcal{S} \subset \mathcal{T}$

$$\left[\begin{array}{cccc|cccc} 1&1&1&1&1&1&0&0 \\ 1&1&1&1&1&1&0&0 \\ 0&0&0&0&1&1&0&0 \\ 0&0&0&0&1&1&0&0 \end{array}\right] \in \mathcal{S}$$



All elements of $\mathcal{S}$ commute with all elements of $\mathcal{T}$

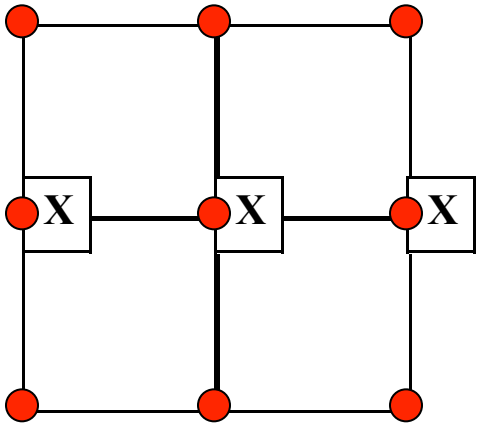
X Columns, Z Rows

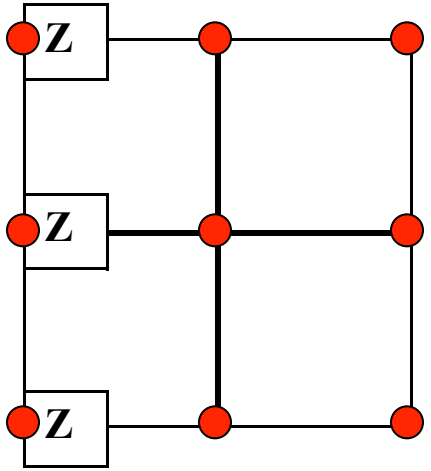
$$\mathcal{L} = \{c\left[\begin{array}{c|c} A & B \end{array}\right] \mid A \text{ has an odd \# rows with } 1\text{'s in entire columns}$$

B has an odd # colums with 1's in entire rows$\}$

$$c\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \in \mathcal{L} \qquad \mathcal{L} \notin \mathcal{T}$$

Elements of $\mathcal{L}$ commute with all elements of $\mathcal{T}$





A Little Representation Theory

$\mathcal{S} = \{ \left[\, A \, \middle| \, B \, \right] \;\; | \;\; A$ has an even # rows with 1's in entire row

B has an even # columns with 1's in entire column$\}$

$\mathcal{S}$ commutes with all of $\mathcal{T}$ and $\mathcal{L}$

Label subspaces by $S = \{ \pm 1$ eigenvalues of elements of $\mathcal{S} \}$

$$\mathcal{H} = \bigoplus_{s \in S} \mathcal{H}_T \otimes \mathcal{H}_L$$

$\dim = 2^{(n-1)^2}$ $\qquad$ $\dim = 2$

Elements $T \in \mathcal{T}$ act as

$$T = \bigoplus_{s \in S} T_s \otimes I$$

Elements $L \in \mathcal{L}$ act as

$$L = \bigoplus_{s \in S} I \otimes L_s$$

Elements $S \in \mathcal{S}$ act as

$$S = \bigoplus_{s \in S} \pm I \otimes I$$

We will encode a single qubit of in $\mathcal{H}_L$

Quantum Error Correcting

$$\mathbf{S}\mathcal{H} = \bigoplus_{s \in S} \mathcal{H}_T \otimes \mathcal{H}_L \mathbf{s}$$

Initialize state over subspace labeled by all $s = +1$ eigenvalues of $S \in \mathcal{S}$ as

$$\rho_{arb} \otimes |\psi\rangle\langle\psi|, \; |\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$$

Error correction for error E need only protect subsystem

$\mathcal{S} = \{\left[\begin{array}{c|c} A & B \end{array}\right] \;|\; A$ has an even # rows with $1'$s in entire row

B has an even # columns with $1'$s in entire columns$\}$

$\mathcal{S}_X = \{\left[\begin{array}{c|c} A & B \end{array}\right] \in \mathcal{S}|$ adjacent rows of A are $1'$s$\}$

$\mathcal{S}_Z = \{\left[\begin{array}{c|c} A & B \end{array}\right] \in \mathcal{S}|$ adjacent columns of B are $1'$s$\}$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \in \mathcal{S}_X \qquad \left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{array}\right] \in \mathcal{S}_Z$$

Diagnose and Pseudo-

1 Initially in $+1$ eigenstate of elements of $\mathcal{S}_X$ and $\mathcal{S}_Z$

2 Error occurs

3 Measure elements of $\mathcal{S}_X$ and $\mathcal{S}_Z$

4 Eigenvalues of measurements allow diagnosis and fix modulo $\mathcal{T}$

X error:
$$\left[\begin{array}{cccc|cccc} 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0 \end{array}\right]$$

Flips to -1 eigenvalues of
$$\left[\begin{array}{cccc|cccc} 0&0&0&0&0&1&1&0\\ 0&0&0&0&0&1&1&0\\ 0&0&0&0&0&1&1&0\\ 0&0&0&0&0&1&1&0 \end{array}\right]$$

$$\left[\begin{array}{cccc|cccc} 0&0&0&0&1&1&0&0\\ 0&0&0&0&1&1&0&0\\ 0&0&0&0&1&1&0&0\\ 0&0&0&0&1&1&0&0 \end{array}\right]$$

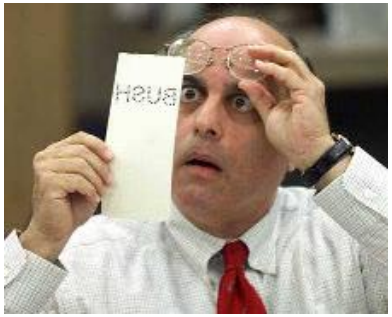
Apply fixing procedure
$$\left[\begin{array}{cccc|cccc} 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0 \end{array}\right]$$

Cumulative error:
$$\left[\begin{array}{cccc|cccc} 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0 \end{array}\right] \in \mathcal{T}$$

$$T = \bigoplus_{s \in S} T_s \otimes I$$

$$\rho_{err} \otimes |\psi\rangle\langle\psi|$$

Detailing The



$\mathcal{T} = \{c\left[\begin{array}{c|c} A & B \end{array}\right]$ | A has an even # of 1′s in columns

B has an even # of 1′s in rows$\}$

violate $\mathcal{T}$ condition

$$\begin{array}{cccc} e & e & o & e \end{array}$$

$$\left[\begin{array}{cccc|cccc} 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} o \\ e \leftarrow \text{violate } \mathcal{T} \text{ condition} \\ e \\ e \end{array}$$

syndrome:

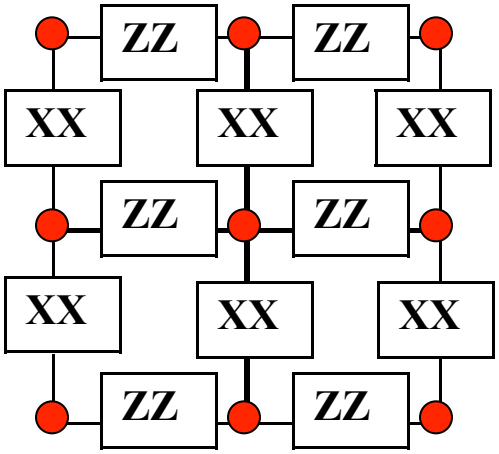
$\mathcal{S}_X = \{\left[\begin{array}{c|c} A & B \end{array}\right] \in \mathcal{S}|$ adjacent rows of A are 1′s$\}$

fix by applying Z's consistent with fewest o's

fixes $\lfloor \frac{n}{2} \rfloor$-1 Z errors

Similar for $\mathcal{S}_Z$

Relation To Physics



Eigenstates of

$$H = \sum_{ij} X_{i,j} X_{i+1,j} + Z_{i,j} Z_{i,j+1}$$

are a quantum error correcting subsystem

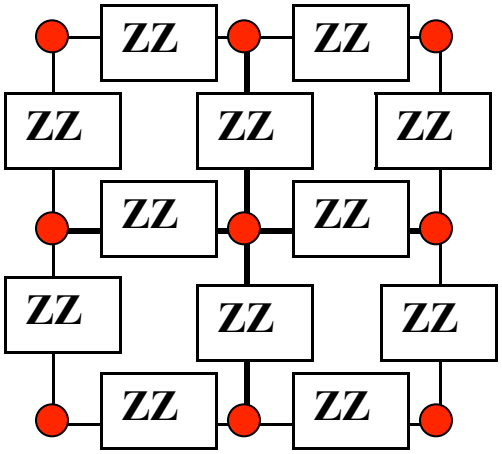
$$H = \bigoplus_{s \in S} H_s \otimes I_2$$

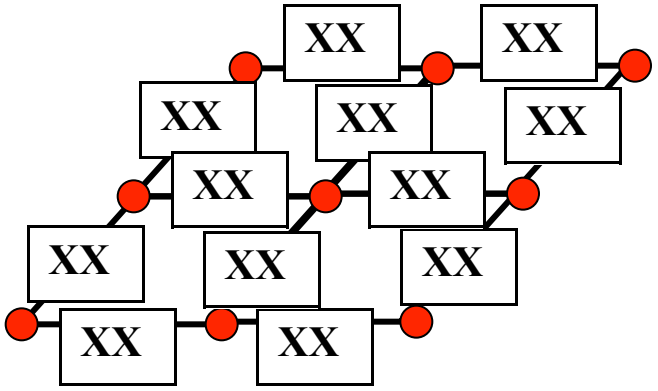
Expect excitations to be pointlike…not self-correcting

Frinkahedro

n by n by cube of qubits







$$\mathcal{T} = \{c\left[\begin{array}{c|c} A & B \end{array}\right] \mid A \text{ has an even \# of } 1\text{'s in YZ plane}$$
$$B \text{ has an even \# of } 1\text{'s in XY plane}\}$$

$$\mathcal{S} = \{\left[\begin{array}{c|c} A & B \end{array}\right] \mid A \text{ has an even \# XY planes with } 1\text{'s in entire plane}$$
$$B \text{ has an even \# YZ planes with } 1\text{'s in entire plane}\}$$

$$\mathcal{L} = \{\left[\begin{array}{c|c} A & B \end{array}\right] \mid A \text{ has an odd \# XY planes with } 1\text{'s in entire plane}$$
$$B \text{ has an odd \# YZ planes with } 1\text{'s in entire plane}\}$$

Expectation s

$$H = \sum_{ijk} X_{i,j,k}X_{i+1,j,k} + X_{i,j,k}X_{i,j+1,k} + Z_{i,j,k}Z_{i,j+1,k} + Z_{i,j,k}Z_{i,j,k+1}$$

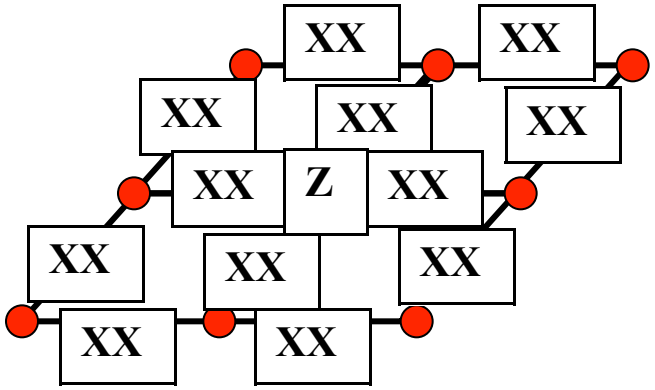
eigenstates are quantum error correcting subsystem

assume phase with two point n.n. correlations

$$\langle X_{i,j,k}X_{i+1,j,k}\rangle = -m_{xx}^{(x)} \qquad \langle Z_{i,j,k}Z_{i,j+1,k}\rangle = -m_{zz}^{(y)}$$

$$\langle X_{i,j,k}X_{i,j+1,k}\rangle = -m_{xx}^{(y)} \qquad \langle Z_{i,j,k}Z_{i,j,k+1}\rangle = -m_{zz}^{(z)}$$

$$\langle H\rangle = -n^2(n-1)(m_{xx}^{(x)} + m_{xx}^{(y)} + m_{zz}^{(y)} + m_{zz}^{(z)})$$



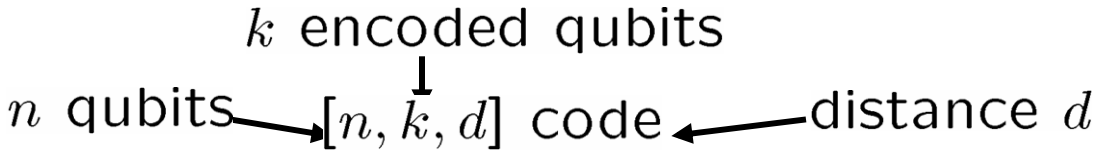
errors are stringlike excitations
in this “mean field” approximation

Energy proportional to perimeter
self correcting?

$$\langle H\rangle_{er} = -n^2(n-1)(m_{xx}^{(x)} + m_{xx}^{(y)} + m_{zz}^{(y)} + m_{zz}^{(z)}) + 2m_{xx}^{(x)} + 2m_{xx}^{(y)}$$

Mean Field Correct?

"To be an Error and to be Cast out is Part of [Nature's] Design."
-*William Blake*



[4,2,2] quantum error detecting code:

$$\mathbf{S}_1 = \mathbf{XXXX} \quad \mathbf{S}_2 = \mathbf{ZZZZ} \qquad \mathbf{S}_i|\psi\rangle = |\psi\rangle$$

Encoded operations on encoded ($\mathbf{S}_i|\psi\rangle = |\psi\rangle$) two qubits

$$\mathbf{X}_1 = \mathbf{XXII},\ \mathbf{Z}_1 = \mathbf{IZZI}$$

$$\mathbf{X}_2 = \mathbf{IXXI},\ \mathbf{Z}_2 = \mathbf{ZZII}$$

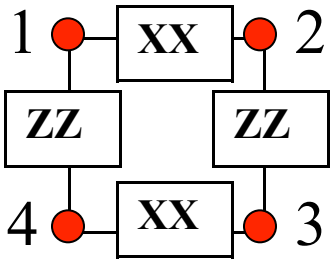
Complete basis: $|S_1, S_2, Z_1, Z_2\rangle,\ S_i, Z_i \in \{\pm 1\}$

$$\mathbf{P}\mathbf{S}_i = -\mathbf{S}_i\mathbf{P} \text{ for some } i$$

$$\langle i_L|\mathbf{P}|j_L\rangle = 0,\ \mathbf{P} \text{ single qubit operator}$$

Supercoherenc

4 qubit system



ϵ

$$S_1 = XXXX$$
$$S_2 = ZZZZ$$
$$X_1 = XXII,\ Z_1 = IZZI$$
$$X_2 = IXXI,\ Z_2 = ZZII$$

System Hamiltonian:

$$H = XXII + IIXX + ZIIZ + IZZI$$

Note symmetries: $[H, S_i] = [H, X_2] = [H, Z_2] = 0$

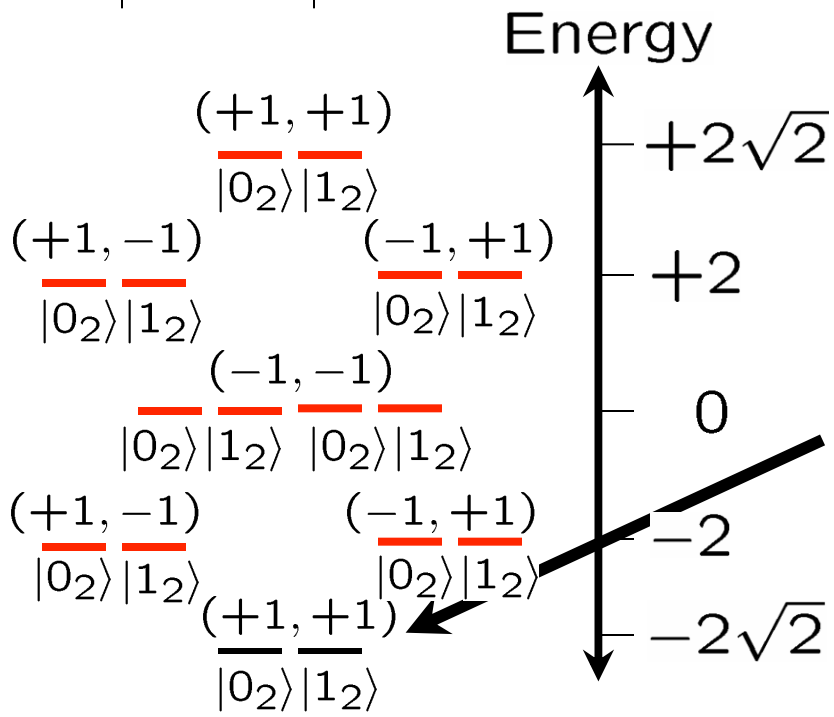
$$H = X_1(I + S_1) + Z_1(I + S_2)$$

Does not involve 2nd encoded qubit.

$\Downarrow$

every energy level at least two-fold degnerate

S_1	S_2	$\mathbf{H} = \mathbf{X}_1(1+S_1) + \mathbf{Z}_1(1+S_2)$	Eigenvalues
$+1$	$+1$	$2(\mathbf{X}_1 + \mathbf{Z}_1)$	$\pm 2\sqrt{2}$
$+1$	-1	$2\mathbf{X}_1$	± 2
-1	$+1$	$2\mathbf{Z}_1$	± 2
-1	-1	$\mathbf{0}$	0



(S_1, S_2) label stabilizer

$|0_2\rangle$, $|1_2\rangle$ second encoded qubit

Degenerate ground state is error detecting state

Bond expectation values are finite and negative.

Supercoherenc

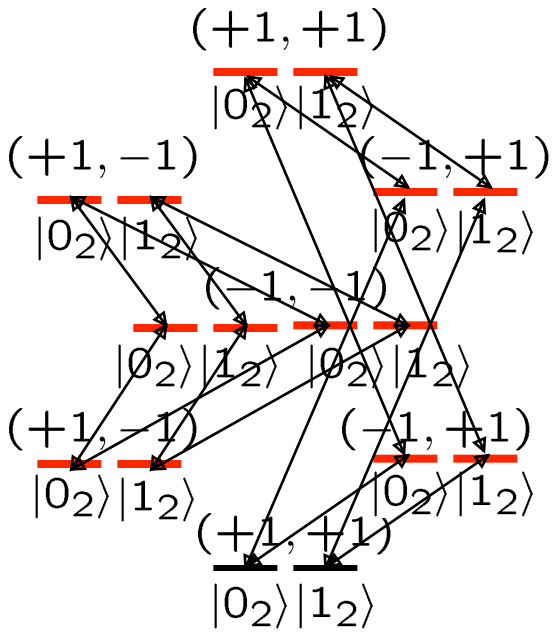
$$\mathbf{H} = \mathbf{X}_1(1 + S_1) + \mathbf{Z}_1(1 + S_2)$$

$\mathbf{S}_i\mathbf{P} = -\mathbf{P}\mathbf{S}_i$ for some i $\quad$ $\mathbf{P}$ single qubit operator

ALL single qubit errors take degenerate ground state to higher energy levels.

Single qubit errors change value of (S1,S2) and hence take ground state to higher energy level.

Effect of **ZIII**

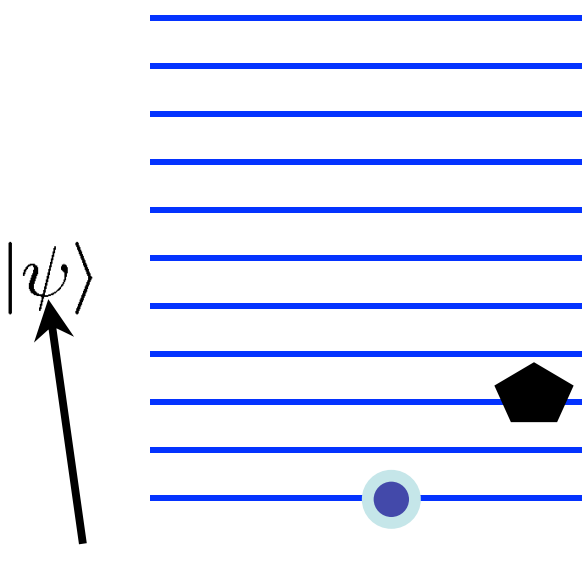


$$\underbrace{(+1, +1)}_{|0_2\rangle \qquad |1_2\rangle}$$

✗ **FORBIDDEN**

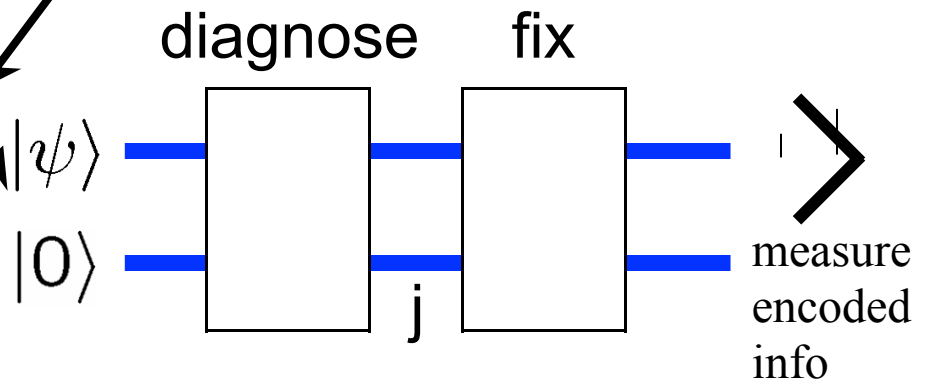
D. Bacon, Ph.D. thesis, U.C. Berkeley, 2001

Quantum Error Correcting Order Parameter



ρ_{error}

The quantum information is still here. How do we see it?



Encoded quantum information

$$|\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$$

$$\mathbf{Q}_s = \sum_j \mathbf{L}_s \mathbf{F}_j \mathbf{D}_j$$

$\mathbf{L}_s$ logical encoded operation

$\mathbf{D}_j$ projector onto jth errored subspace

$\mathbf{F}_j$ fixing operator for jth errored subspace

Subsystem Order Parameter

$$\mathcal{T} = \{c\left[\begin{array}{c|c} A & B \end{array}\right] \;\mid\; A \text{ has an even \# of } 1\text{'s in columns}$$
$$B \text{ has an even \# of } 1\text{'s in rows}\}$$

violate $\mathcal{T}$ condition

$$\begin{array}{cccc} e & e & o & e \end{array}$$
$$\left[\begin{array}{cccc|cccc} 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} o \\ e \leftarrow \text{violate } \mathcal{T} \text{ condition} \\ e \\ e \end{array}$$

Suppose probability of error per qubit is p.

If p<1/(2L), then error correction will succeed.

Contrast with toric codes percolation threshold.

Finite Size Saves

Error rate scales like:

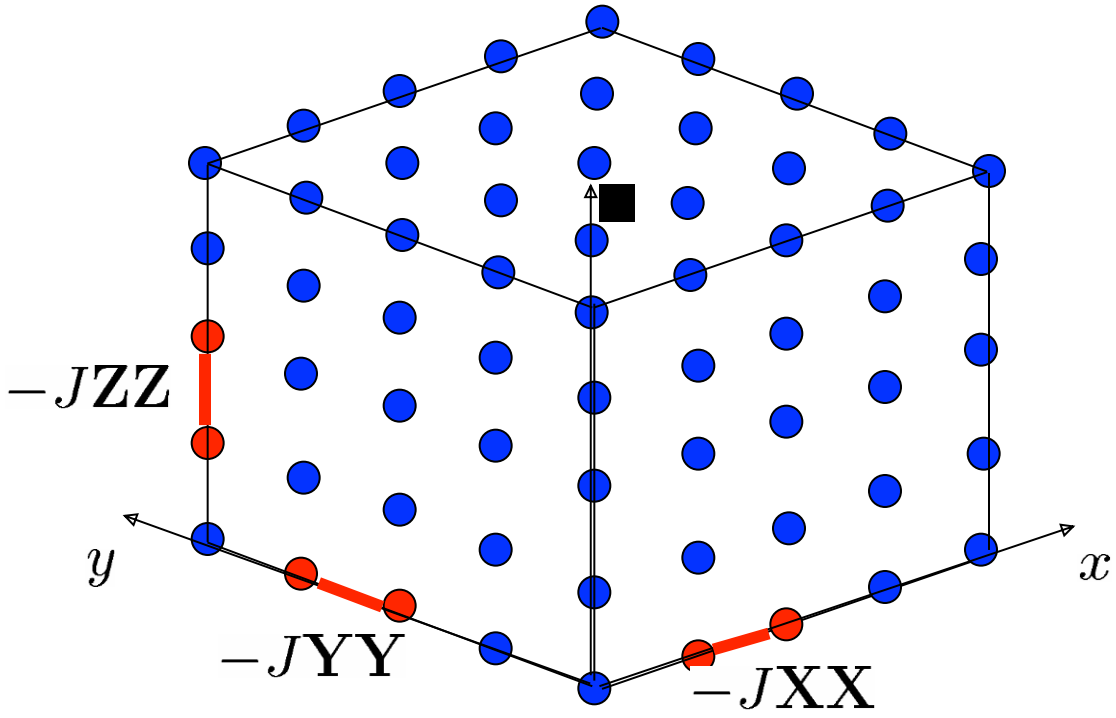
$$\exp\left(-\frac{\Delta}{kT}\right)$$

This must be

$$\exp\left(-\frac{\Delta}{kT}\right) \approx \frac{1}{L}$$

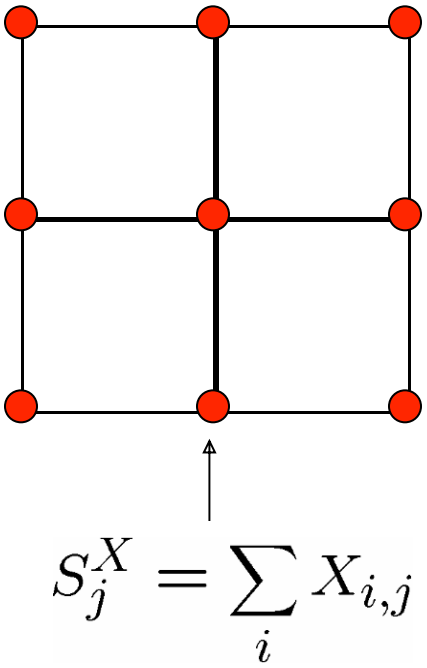
The system must be gapped in order for the subsystem code to allow readout of encoded quantum information.

Back to the Third Dimension



$$\mathbf{H} = -J \sum_{ijk} (\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{Y}_{i,j,k}\mathbf{Y}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1})$$

Long Range Interactions?



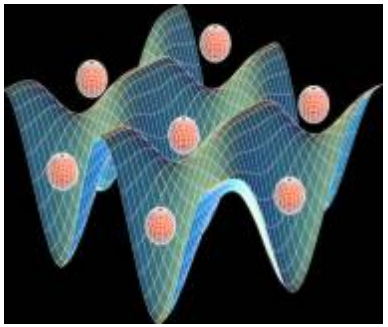
$$S_i^Z = \sum_j Z_{i,j}$$

$$S_j^X = \sum_i X_{i,j}$$

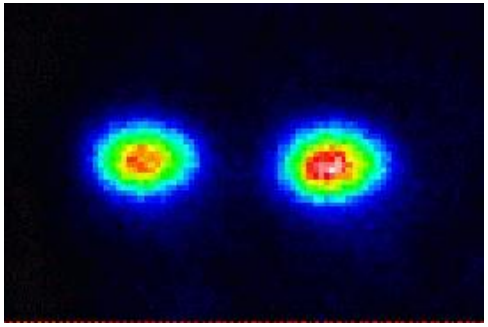
$$H = -J\left(\sum_j (S_j^X)^2 + \sum_i (S_i^Z)^2\right)$$

Same 2D subsystem code, but now mean field argument for the Hamiltonian yields self-correction (Weiss-Ising)

Physical Implementations?



optical lattice



ion traps?

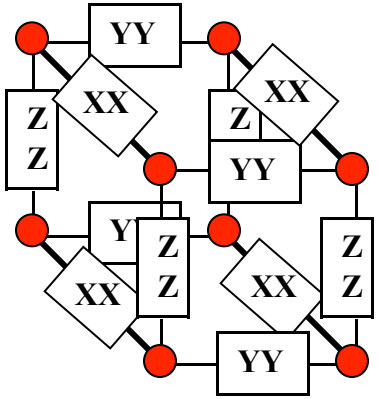
Quantum Computing



IST Boulder Ions

Self Correcting Quantum Computers

quantum error correcting subsystems in spin lattice systems



self-correcting quantum memories?